

Spring Boot Backend

Transit KMS Client Service

- Endpoint: `http://haproxy:8200`
- AES Key Wrapping & Unwrapping
- KMS Shredding & RSA-2048 Proof Signing
- Tombstone Reconciler & Memory Zeroize

HAProxy Load Balancer

Port :8200 (Single Gateway Entrypoint)

- Health Poller: `GET /v1/sys/health` (500ms)
- HTTP 200 OK → Route to Active Leader
- HTTP 429 → Bypass Standby Followers
- Zero-Downtime Auto-Failover (< 2.5s)

HashiCorp Vault 3-Node Raft Cluster

Integrated Storage WAL | Quorum: 2/3 Nodes

Vault Node 2 (vault-2)

STANDBY

Hot Standby Follower (:8200)

- Raft Quorum Voter | Append-Only WAL Sync
- Health: HTTP 429 (Standby Replica)

Vault Node 3 (vault-3)

STANDBY

Hot Standby Follower (:8200)

- Raft Quorum Voter | Append-Only WAL Sync
- Health: HTTP 429 (Standby Replica)

Vault Node 1 (vault-1)

ACTIVE LEADER

Active Raft Leader (:8200)

- Transit Secrets Engine Mounted
- Master KEK Lifecycle & Destruction
- RSA-2048 Proof Signing Authority
- Health: HTTP 200 OK (Active Leader)
- Direct Master Key Wrap/Unwrap

Raft Linearizable WAL Replication — Heartbeats 250ms | AppendEntries :8201

Automated Disaster Recovery & Failover Workflow (< 2.5s)

1. Heartbeat Loss

vault-1 crashes.
Followers detect
timeout (>500ms).

2. Raft Election

vault-2 starts T+1.
Secures 2/3 quorum,
re-elects in 1.84s.

3. State Promotion

vault-2 mounts
transit engine,
returns HTTP 200.

4. Traffic Reroute

HAProxy detects
health in 2.15s,
failover < 2.5s.